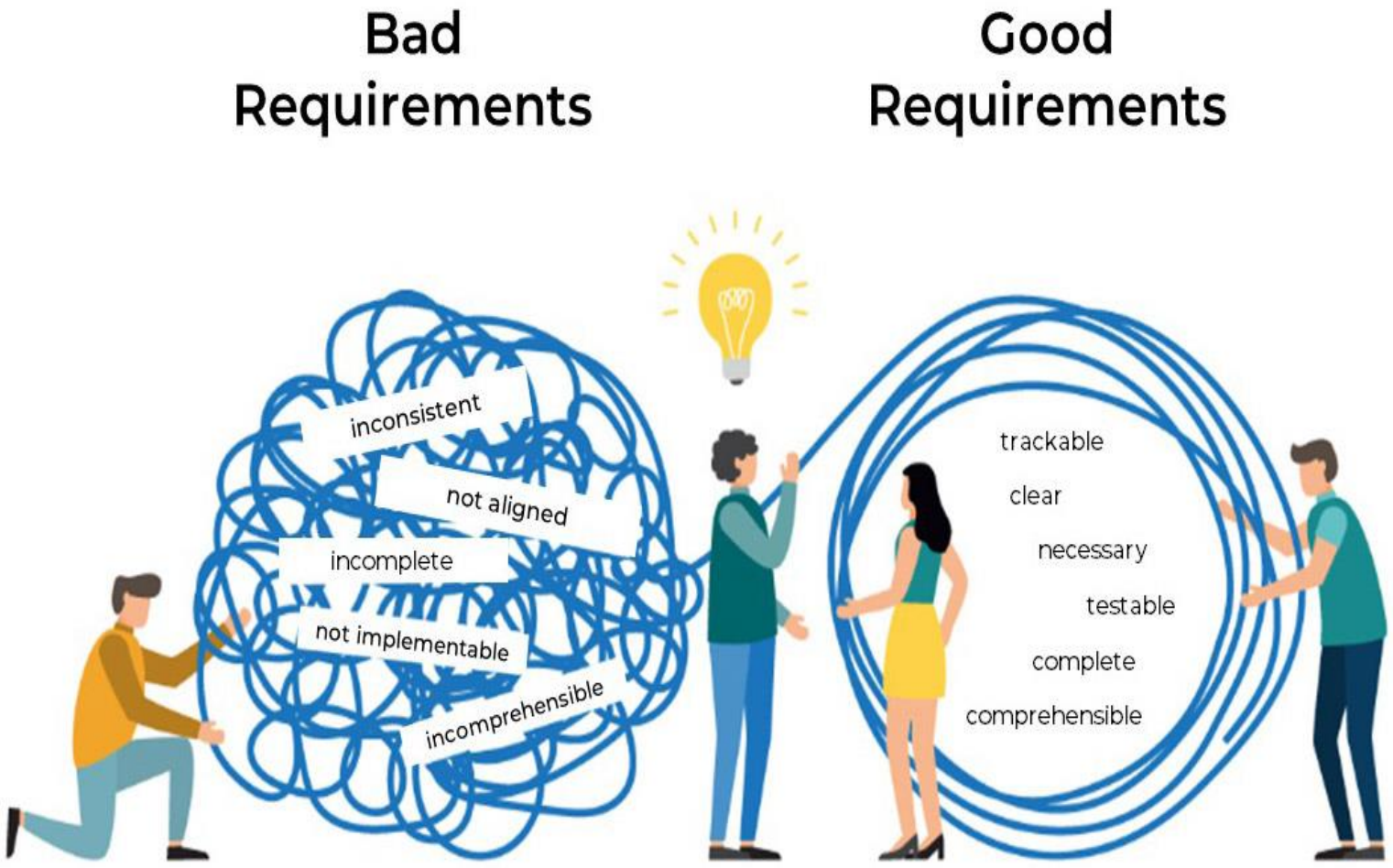
SCS1207
Software Engineering I

H N D THILINI

Quiz Time

Password : Ag#12$04

Requirement Engineering



Topics Covered

- What is a Requirement?
- Functional and Non-functional Requirements
- Requirements Engineering Processes
- Requirements Elicitation and Analysis
- Requirements Specification
- Requirements Validation
- Requirements Change

What is a requirement?

Activity 1

- List down a few requirements of your daily life?

What is a requirement?

- Requirements reflect the needs of customers for a system that serves a certain purpose such as controlling a device, placing an order, or finding information.
- Range from a high-level abstract statement of a service/ constraint to a detailed mathematical functional specification.
- The process of finding out, analyzing, documenting and checking these services and constraints is called requirements engineering (RE).

Requirement Engineering

The process of gathering and defining of what services should be provided by the system

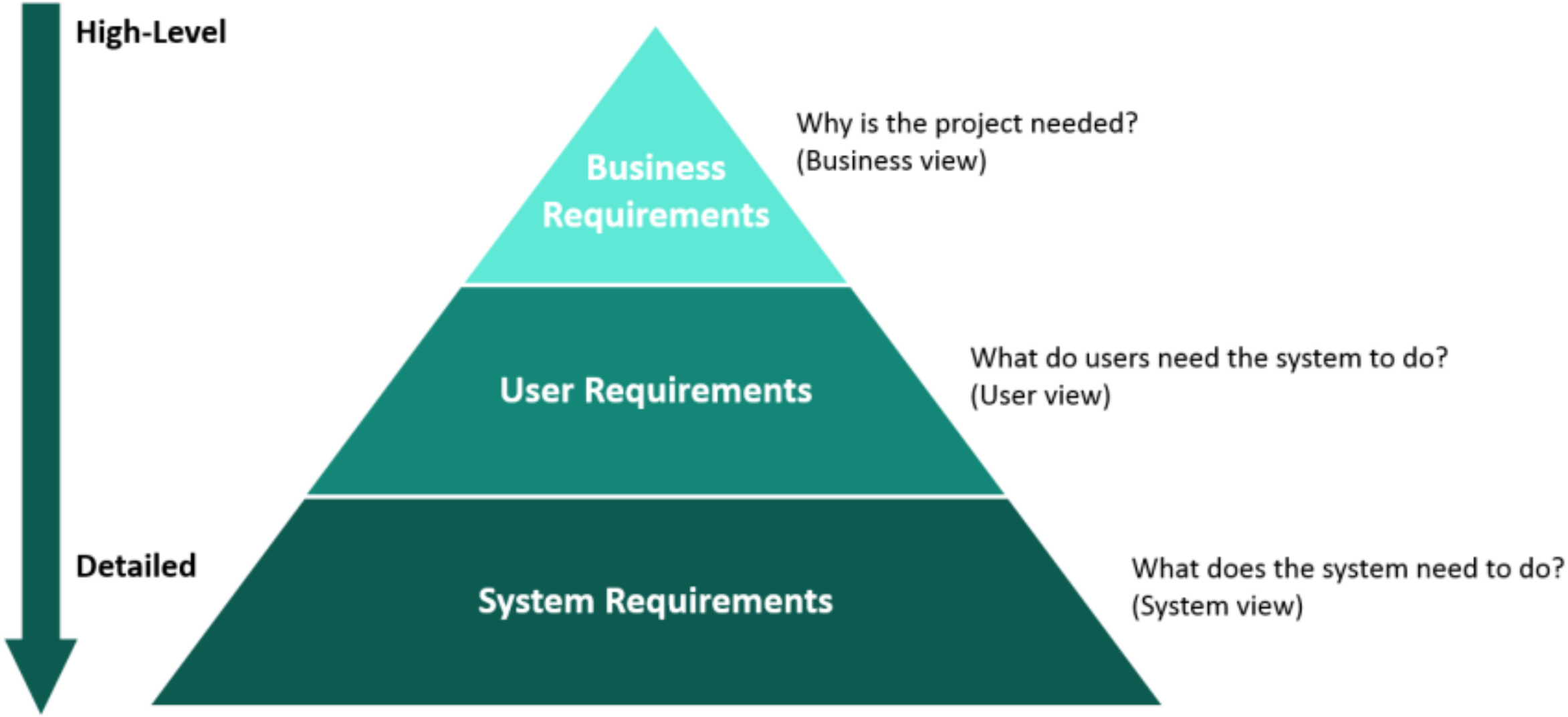
and

the constraints under which it operates and is developed

Requirements abstraction (Davis, A Researcher)

"If a company wishes to let a contract for a large software development project, it must **define its needs in a sufficiently abstract way** that a solution is not pre-defined. The requirements must be written so that **several contractors can bid** for the contract, offering, perhaps, **different ways of meeting the client organization's needs**. Once a contract has been awarded, the **contractor must write a system definition for the client in more detail** so that the **client understands and can validate what the software will do**. Both of these documents may be called the **requirements document for the system**."

Types of Requirements



Types of Requirements

- User requirements

  Statements in natural language plus **diagrams** of the ***services the system provides and its operational constraints***. Written for customers.

- System requirements

  A **structured document** setting out **detailed descriptions of the system's functions, services and operational constraints**. Defines **what should be implemented** so may be part of a contract between client and contractor.

Types of Requirements

- **User requirements** are **requirements** that have been communicated by the **user**.

  "I expect the **system** to do X"

- **System requirements** are **requirements** imposed by the design of the **system** itself.

  "Because the **user** has to do X, the **system** must do Y."

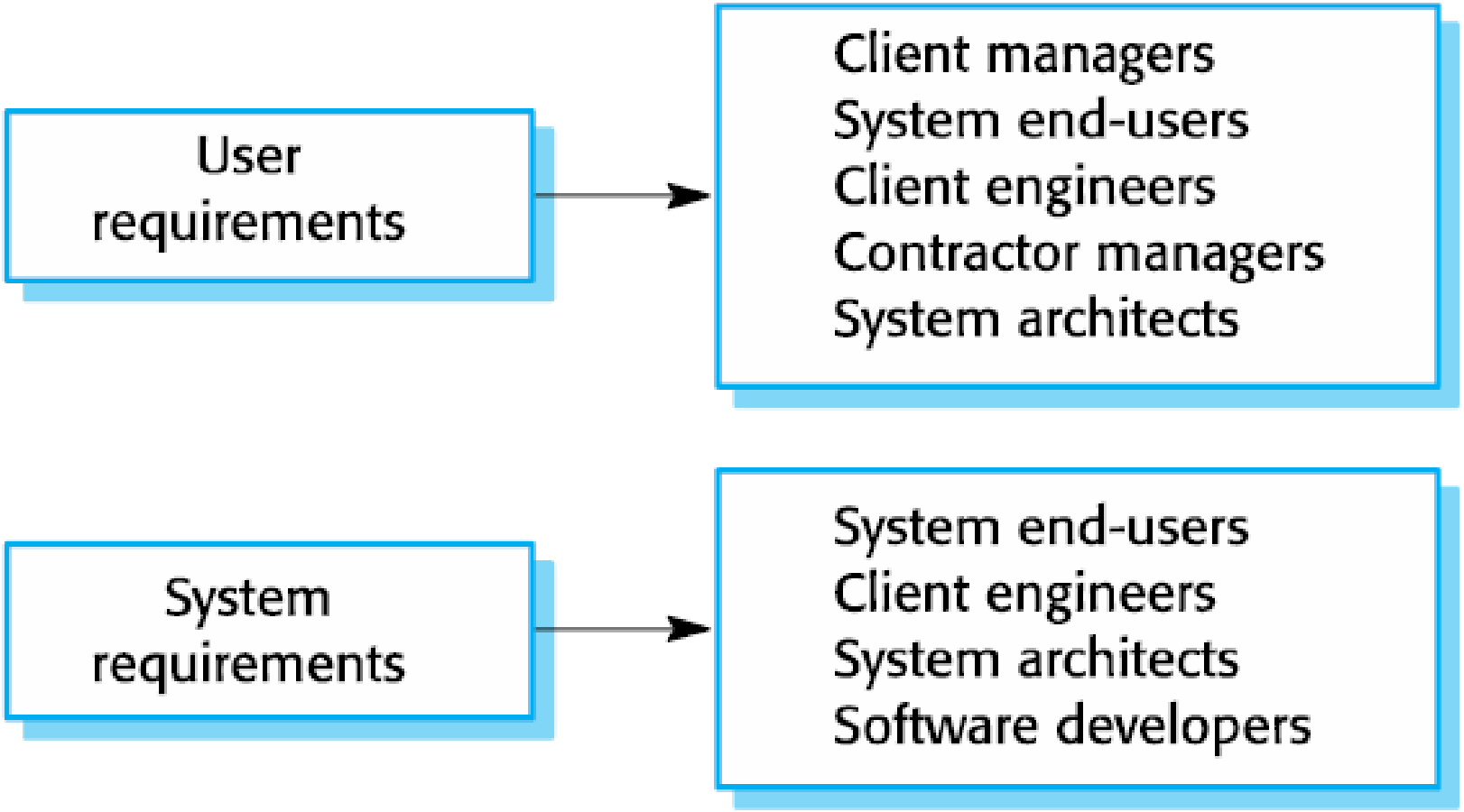
User & System Requirements

1. The Mentca
showing th

System requirements specification

1.1 On the last working day of each month, a summary of the drugs prescribed, their cost and the prescribing clinics shall be generated.
1.2 The system shall generate the report for printing after 17.30 on the last working day of the month.
1.3 A report shall be created for each clinic and shall list the individual drug names, the total number of prescriptions, the number of doses prescribed and the total cost of the prescribed drugs.
1.4 If drugs are available in different dose units (e.g. 10mg, 20mg, etc) separate reports shall be created for each dose unit.
1.5 Access to drug cost reports shall be restricted to authorized users as listed on a management access control list.

Readers of different types of requirements specification



Project/ System stakeholders

- Any **individual** or an **organization** that is actively involved in a project, or whose interest might be affected (positively or negatively) as a result of project execution or completion.

Project/ System stakeholders

- Stakeholder types
  - The project manager
  - The project team
  - System managers / The client
  - System owners
  - End users
  - External stakeholders
    - The project sponsor
    - The external organizations
  - The rest: anyone who might be affected by the project outputs

CASE STUDY: A patient information system for mental health care

- A patient information system to support mental health care is a medical information system that maintains information about patients suffering from mental health problems and the treatments that they have received.
- Most mental health patients do not require dedicated hospital treatment but need to attend specialist clinics regularly where they can meet a doctor who has detailed knowledge of their problems.
- To make it easier for patients to attend, these clinics are not just run in hospitals. They may also be held in local medical practices or community centers.
- The MHC-PMS (Mental Health Care-Patient Management System) is an information system that is intended for use in clinics.
- It makes use of a centralized database of patient information but has also been designed to run on a PC, so that it may be accessed and used from sites that do not have secure network connectivity.
- When the local systems have secure network access, they use patient information in the database but they can download and use local copies of patient records when they are disconnected.

MHC-PMS goals

- To generate management information that allows health service managers to assess performance against local and government targets.
- To provide medical staff with timely information to support the treatment of patients.

Activity 2

- Identify the stakeholders of the system.

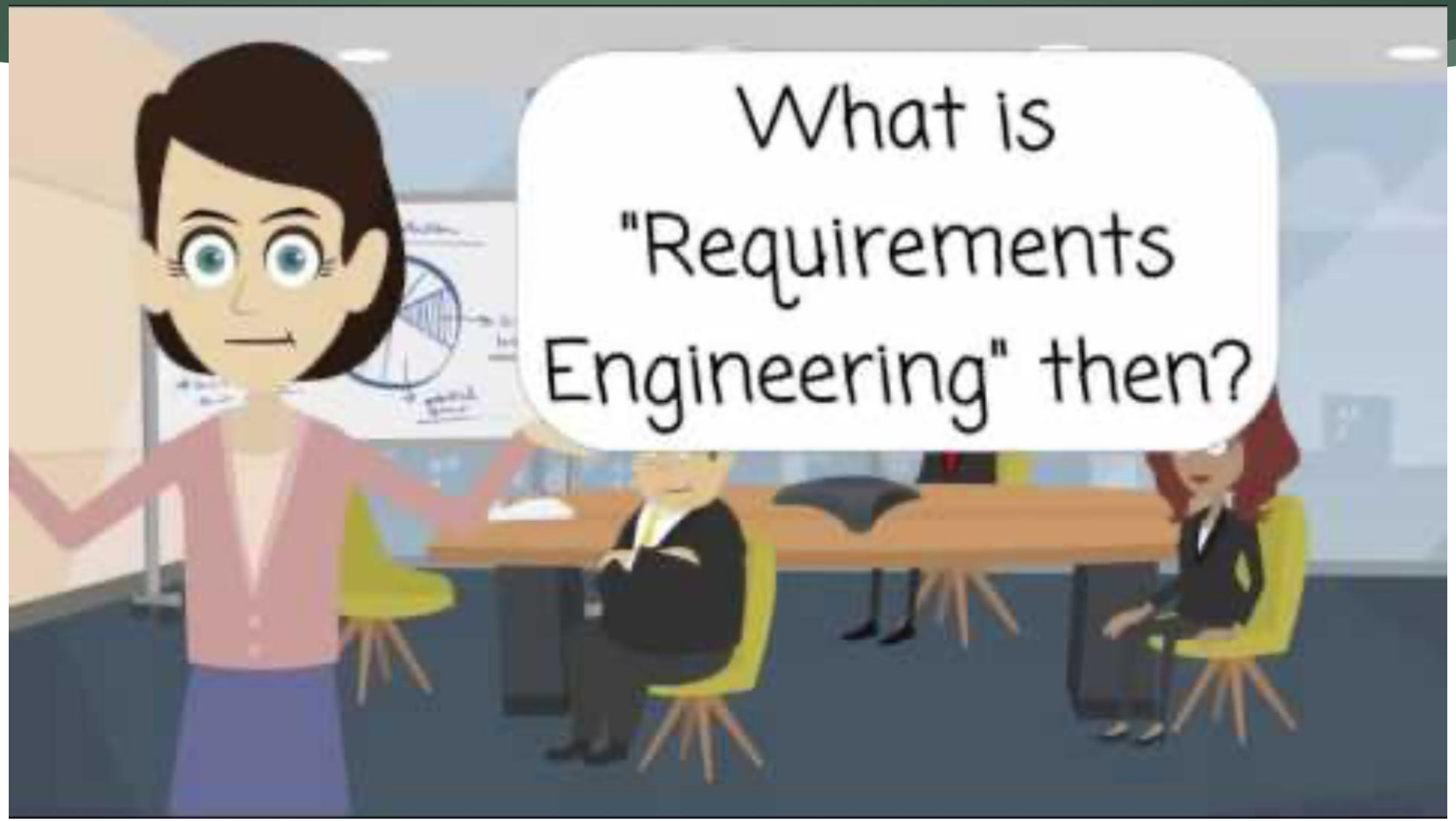
Stakeholders in the MHC-PMS

- **Patients** whose information is recorded in the system.
- **Doctors** who are responsible for assessing and treating patients.
- **Nurses** who coordinate the consultations with doctors and administer some treatments.
- **Medical receptionists** who manage patients' appointments.
- **IT staff** who are responsible for installing and maintaining the system.

Stakeholders in the MHC-PMS

- **A medical ethics manager** who must ensure that the system meets current ethical guidelines for patient care.
- **Health care managers** who obtain management information from the system.
- **Medical records staff** who are responsible for ensuring that system information can be maintained and preserved, and that record keeping procedures have been properly implemented.

Requirements Engineering cont….



Activity 3

Define the following terms:

(a) Requirement

(b) Stakeholder

(c) Requirements Engineering

Define the following terms

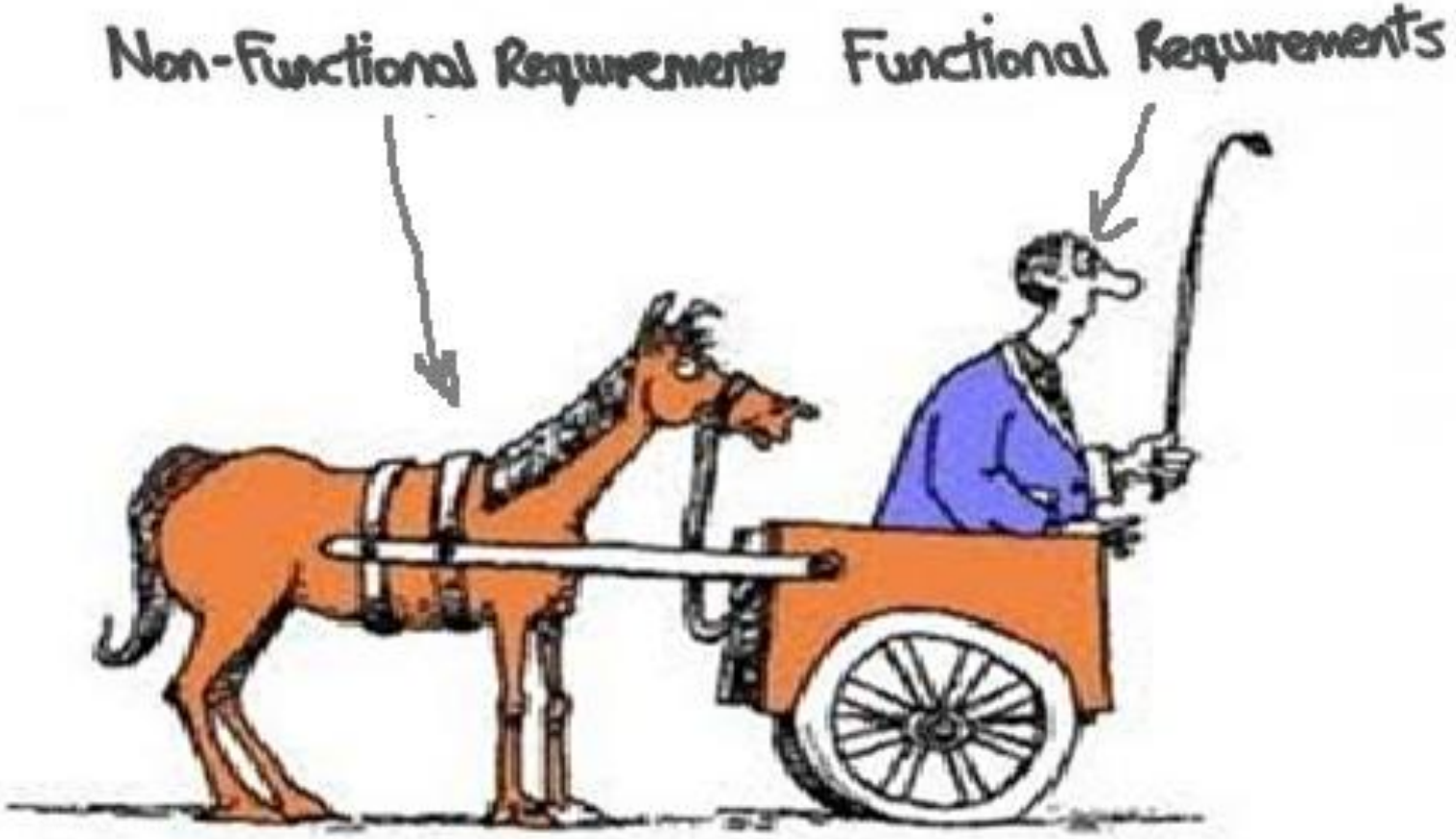
- Requirement
  - Condition or a capability needed by the users to solve a problem or achieve an objective.
- Stakeholder
  - Person / organization that has a direct / indirect influence on the requirements of the system.
- Requirements Engineering
  - Process of capturing stakeholder needs in a way that minimize the risk of delivering a system that does not meet their desires and needs

Agile methods vs Requirements

- Many agile methods argue that **producing detailed system requirements is a waste** of time as **requirements change so quickly**.
- The requirements document is therefore always **out of date**.
- Agile methods usually **use incremental requirements** engineering and may express requirements as **'user stories'.**
- This is **practical for business systems** but **problematic for systems that require pre-delivery analysis** (e.g. critical systems) or **systems developed by several teams**.

Functional
&
Non-Functional
Requirements

Functional & Non-Functional Requirements

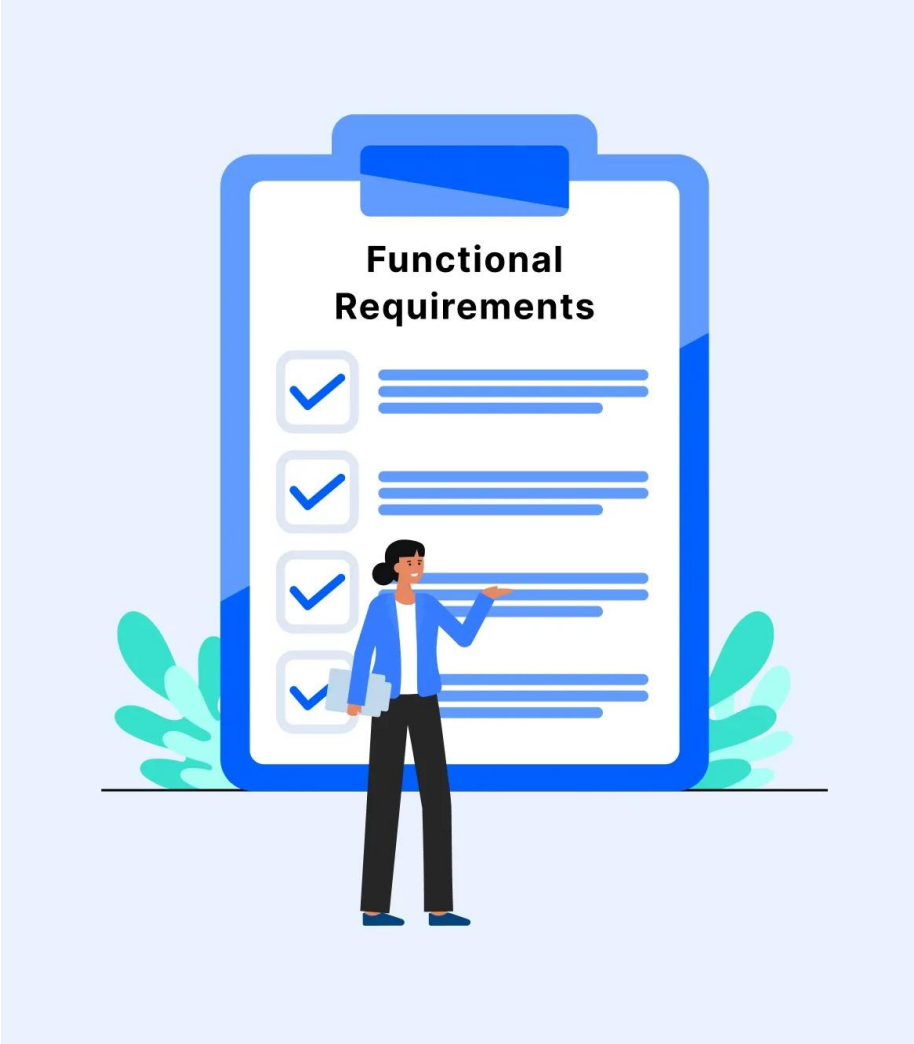


Don't put what you want to do before how you need to do it

Functional & Non-Functional Requirements

- Functional requirements (What the system should do?)
  - Statements of **services** the system should provide,
  - how the system should react to particular inputs and how the system should behave in particular situations.
  - May state what the system should not do.
- Non-functional requirements (How the system performs a certain function?)
  - **Constraints** on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc.
  - Often apply to the system as a whole rather than individual features or services.

Functional Requirements

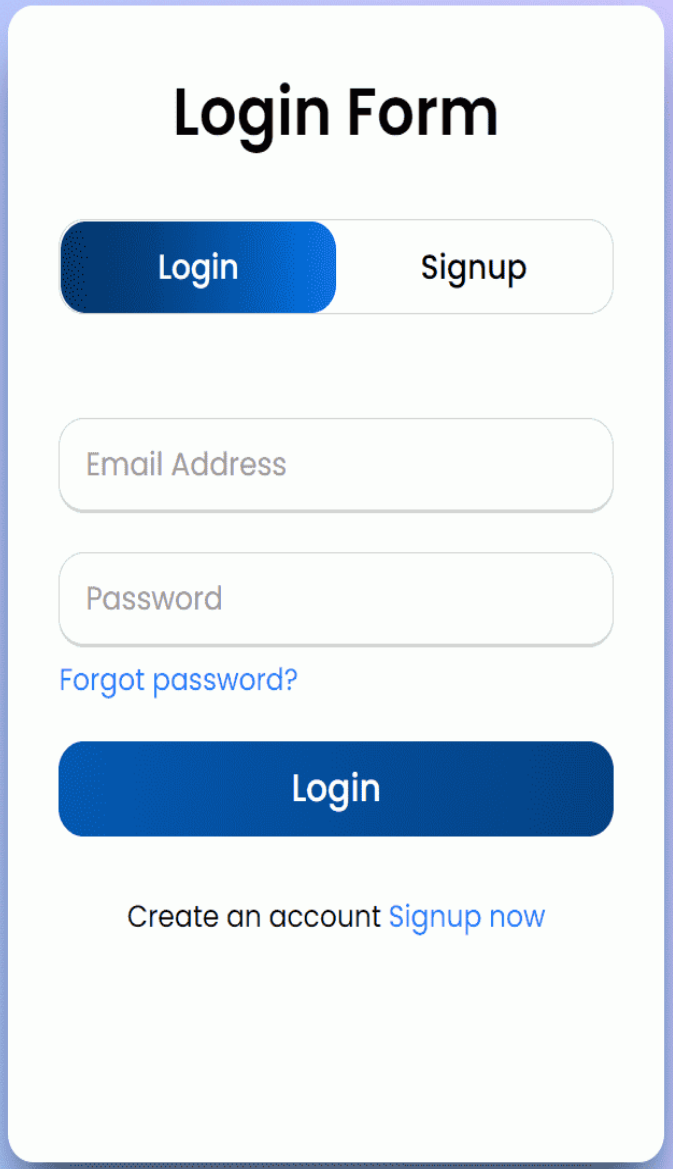


Functional Requirements

- Describe **functionality** or system services.
- Depend on the **type of software**, expected **users** and the type of system **where the software is used**.
- May be high-level statements of what the system should do.
- Can be **written in natural language** so that system users and managers can understand them.
- **Expand the user requirements** and are written for system developers.

Functional Requirements: Example

- *Registered users shall be able to login with valid username and password*
- *On successful login, user shall be redirected to a landing page in the system*
- *On failure, for not registered username prompt "Username not registered" message and for invalid credentials prompt "Invalid credentials"*
- *New users shall be able to register with the system by clicking on the "Sign-Up" link*
- *Users shall be able to recover password by clicking on "Forgot Password" link*



Functional requirements: Mentcare system

"Used to maintain information about patients receiving treatment for mental health problems. "

1. A user shall be able to search the appointments lists for all clinics.
2. The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.
3. Each staff member using the system shall be uniquely identified by his or her 8-digit employee number.

Requirements Imprecision

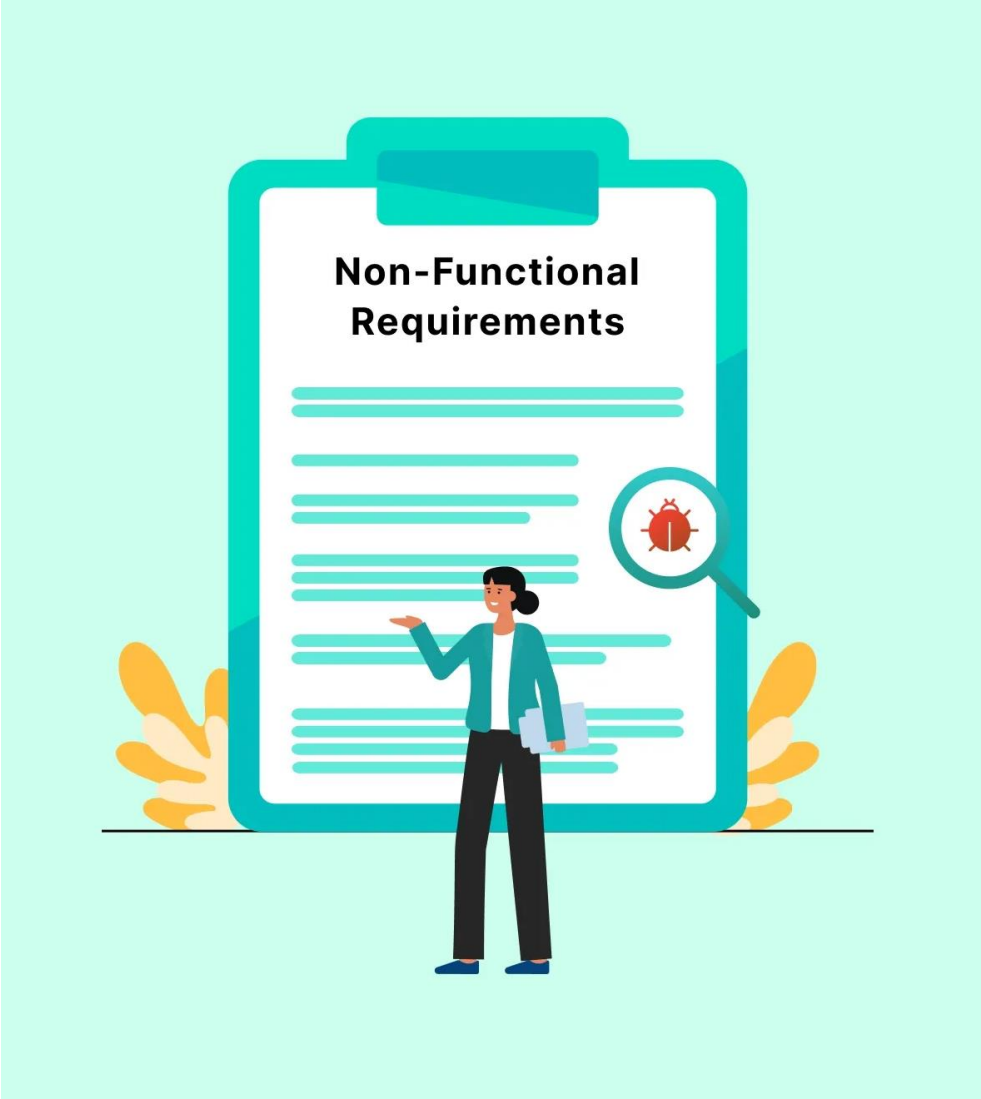
- Problems arise when not precisely stated.
- Ambiguous requirements
  - Can be interpreted in different ways by developers and users.
- The term 'search' in requirement 1
  - User intention –search for a patient name across all appointments in all clinics;
  - Developer interpretation –search for a patient name in an individual clinic. User chooses clinic then search.

"A user shall be able to search the appointments lists for all clinics. "

Requirements completeness and consistency

- Should be both **complete and consistent.**
- Complete
  - They should **include descriptions of all services required**.
- Consistent
  - There should be **no conflicts or contradictions in the descriptions of the system facilities.**
- However…
  - because of system and environmental complexity, it is impossible to produce a complete and consistent requirements document.

Non-Functional Requirements

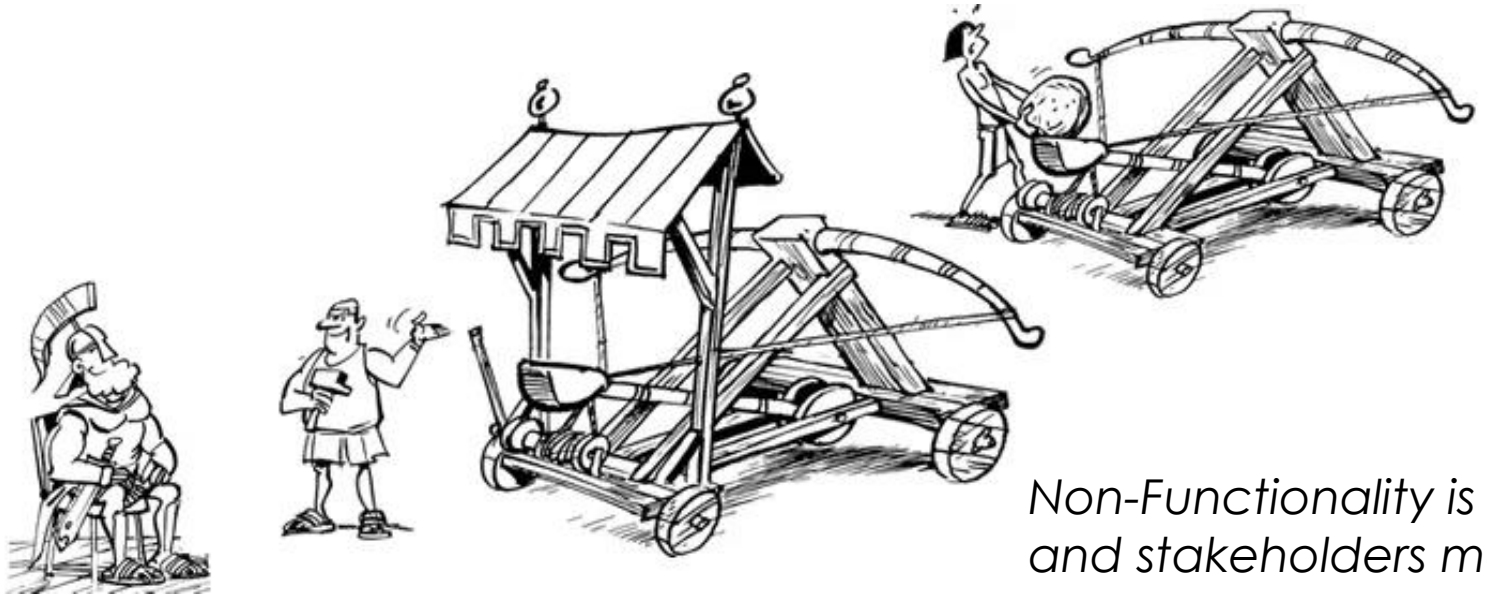


Non-Functional Requirements

- **Requirements** that are **not directly concerned with the specific services delivered by the system to its users**.
- Specify the system‘s quality characteristics‘ or ‘quality attributes‘.
- Specify characteristics of the **system as a whole**.
- Relate to emergent system properties
  - reliability,
  - response time, and
  - memory use.

Non-Functional Requirements

- **Process** requirements may also be specified
  - mandating a particular IDE, programming language or development method.
- Non-functional requirements may be **more critical** than functional requirements. If these are not met, the system may be useless.

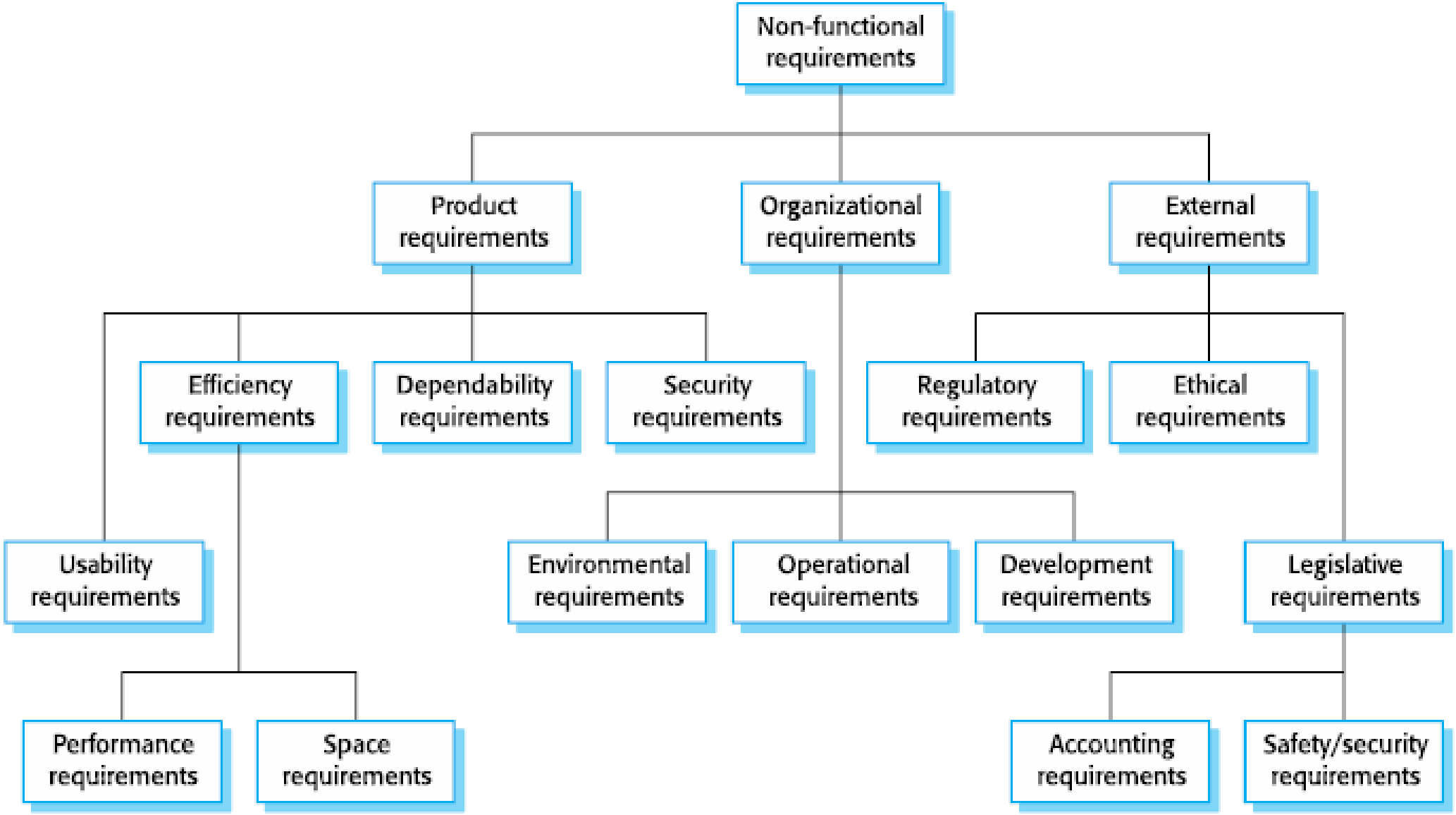


Non-Functionality is in the Eye of the Beholder: users and stakeholders may have different opinions.

Non-Functional Requirements

- May affect the overall architecture of a system rather than the individual components.
  - Example: to ensure that **performance** requirements are met, you may have to organize the system to **minimize communications between components.**
- **NFR** may generate several, related **FR**
  - Example: security requirement might constrain existing FR,
  - may limit access to information in the system

Non-functional requirement classification



Classification of Non-functional Requirements

- *Product requirements*
  - Specify or constrain the **runtime behavior of the software.**
  - e.g. performance requirements for how fast the system must execute and how much memory it requires; reliability requirements that set out the acceptable failure rate; security requirements; and usability requirements.
- *Organizational requirements*
  - Derived from **policies and procedures** in the customer‘s and developer‘s organizations.
  - e.g. operational process requirements that define how the system will be used; development process requirements that specify the programming language.

Classification of Non-functional Requirements

- *External requirements*
  - All requirements that are derived from factors external to the system and its development process
  - may include regulatory requirements that set out what must be done for the system to be approved for use by a regulator, such as a nuclear safety authority
  - legislative requirements that must be followed to ensure that the system operates within the law;
  - ethical requirements that ensure that the system will be acceptable to its users and the general public

Examples of nonfunctional requirements in the Mentcare system

Product requirement

The Mentcare system shall be available to all clinics during normal working hours (Mon–Fri, 0830–17.30). Downtime within normal working hours shall not exceed five seconds in any one day.

Organizational requirement

Users of the Mentcare system shall authenticate themselves using their health authority identity card.

External requirement

The system shall implement patient privacy provisions as set out in HStan-03-2006-priv.

Non-Functional Requirements

- Performance
  - Response Time, Throughput, Utilization, Static Volumetric
- Scalability
- Capacity
- Availability
  - uptime vs downtime
- Reliability
  - software fails under certain conditions
- Recoverability
- Maintainability
- Confidentiality
- Safety
- Data Accuracy and Precision

Non-Functional Requirements

- Interoperability
  - Ability to exchange information and communicate with internal and external applications and systems.
- Portability
  - The effort required to move the software to a different target platform. The measurement is most commonly person-months or % of modules that need changing.
- Usability
  - Requirements about how difficult it will be to learn and operate the system. The requirements are often expressed in learning time or similar metrics

Non-Functional Requirements

- Very **difficult to state precisely** and imprecise requirements may be **difficult to verify**.

✕ **Goal** -Helpful to developers as they convey the intentions of the system users

A general intention of the user such as ease of use.

✓ **Verifiable non-functional requirement**

A statement using some measure that can be objectively tested.

Non-Functional Requirements - Mentcare

- The system should be easy to use by medical staff and should be organized in such a way that user errors are minimized (Goal).

- Medical staff shall be able to use all the system functions after four hours of training. After this training, the average number of errors made by experienced users shall not exceed two per hour of system use (Testable non-functional requirement).

Metrics for specifying nonfunctional requirements

Property	Measure
Speed	Processed transactions/second User/event response time Screen refresh time
Size	Mbytes Number of ROM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems

Group Assignment 03: Identify the requirements of a given scenario

Project Overview:

- A large private hospital wants to implement a **Smart Hospital Management System (SHMS)** to digitally manage patient care, administrative operations, and resource utilization across its departments (e.g., outpatient, inpatient, emergency, diagnostics, pharmacy, billing). The system should streamline workflows for doctors, nurses, administrative staff, lab technicians, and patients.
- The system must also provide real-time dashboards, integrate with third-party devices (like patient monitors), and comply with healthcare regulations.

TASK:

- Identyfy the system ***Stakeholders***
- Identify ***User Requirements***
- Identify the ***System Requirements*** and divide them into ***Functional Requirements*** and ***Non-Functional Requirements***